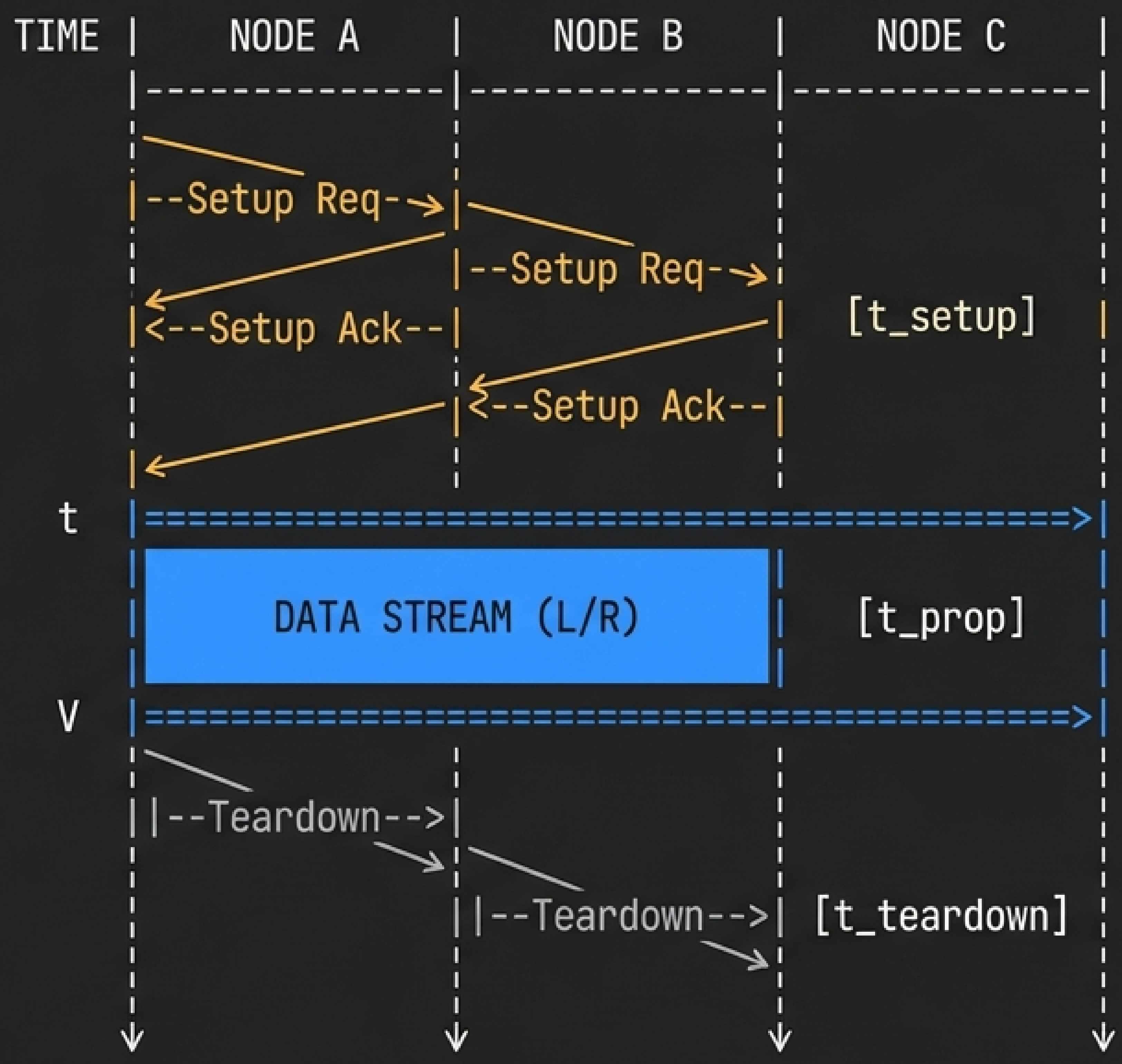
The Anatomy of Circuit Switching

Understand the three-phase state machine and mathematical latency floor of dedicated end-to-end path allocation.

Phase	Network Action	Bandwidth State	Latency Impact
1. Setup	Signaling protocols reserve path	Wasted (idle wait)	High delays
2. Transfer	Bitstream flows	100% Guaranteed	Zero queuing delay
3. Teardown	Memory/fabric release	Unlocking	Negligible

ENGINEER'S LOG

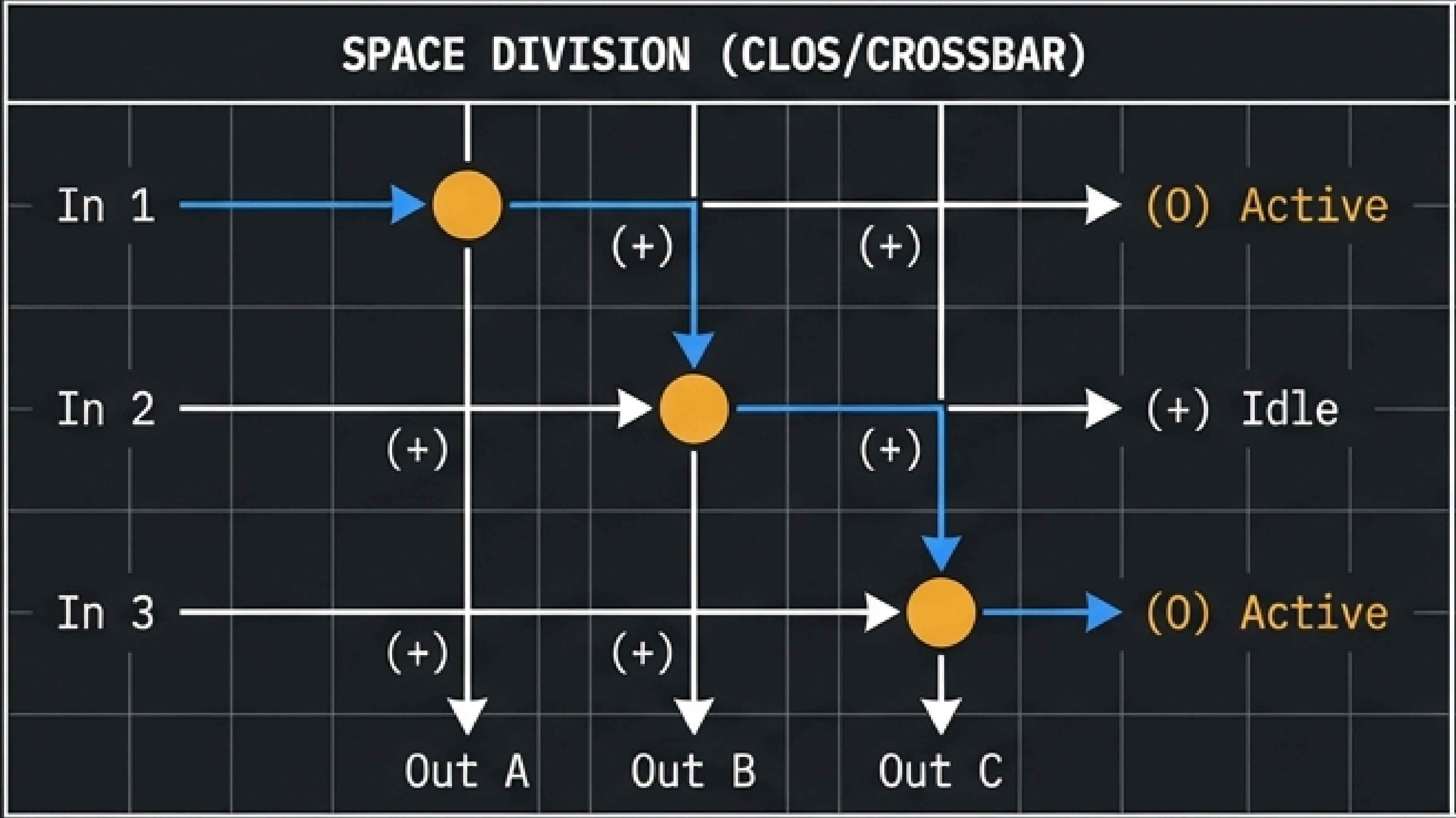
Circuit switching is a stateful architecture with a massive front-end latency penalty during setup. However, transmission time (L/R) only occurs once. Because this circuit acts as a single contiguous pipe, intermediate nodes don't wait for the end of the file to arrive before forwarding. Queuing delay drops to zero—perfect for low-latency streams, but disastrously brittle for bursty data.

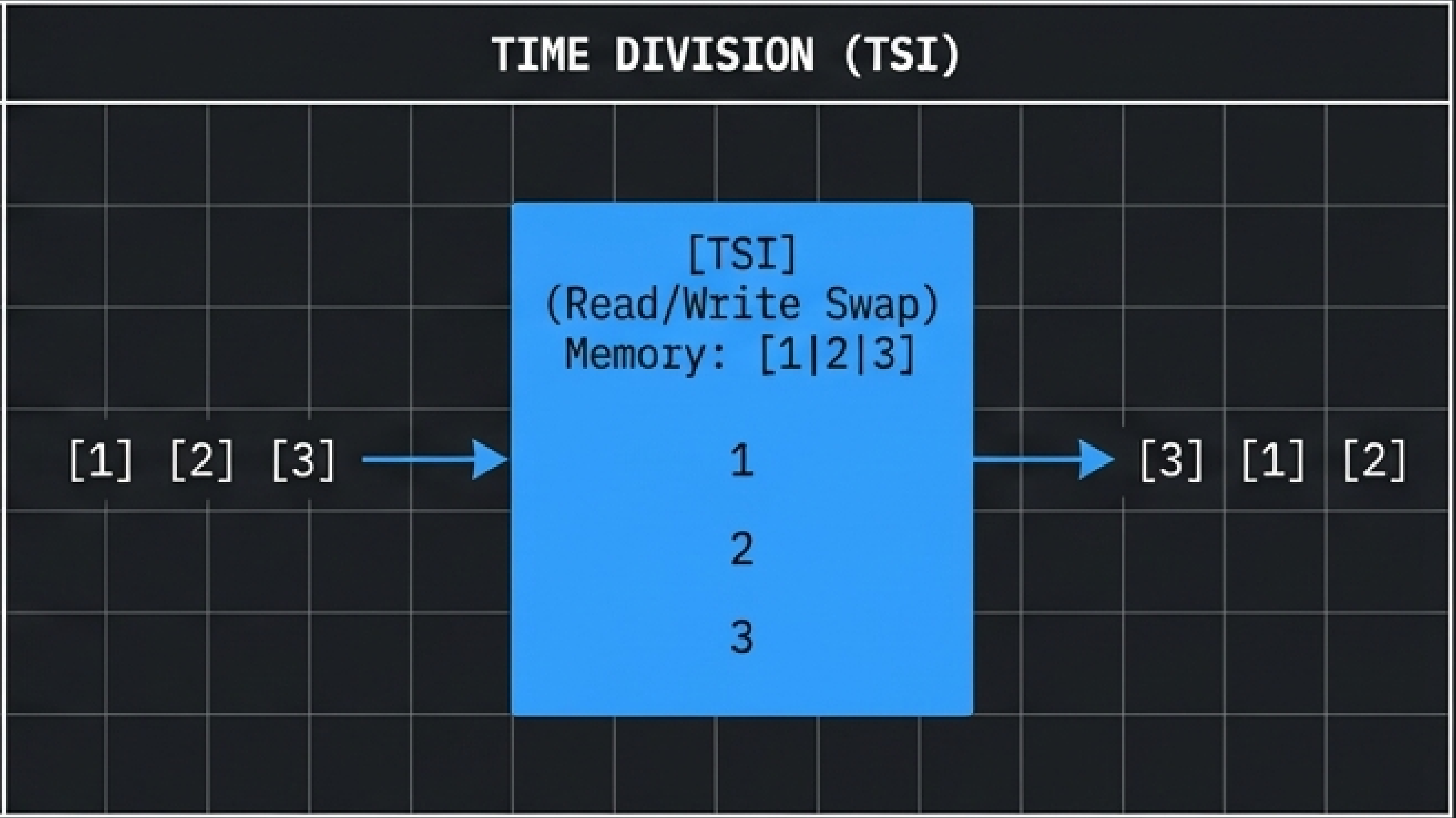


$$T\_total = t\_setup + (L/R) + t\_prop + t\_teardown$$

INTERNAL SWITCH ARCHITECTURE

Differentiate how physical crossbars and memory-based Time Slot Interchangers route data across hardware plains.





Architecture	Mechanism	Constraint	Modern Evolution
Space-Division	Physical matrix of wire crosspoints	Geometric N^2 scaling issue	Multi-stage Clos Networks
Time-Division	Memory read/write sequencing	RAM write speeds	High-speed SRAM fabrics
Hybrid	Space-Time-Space (STS) switching	Synchronization	Carrier-grade core routers

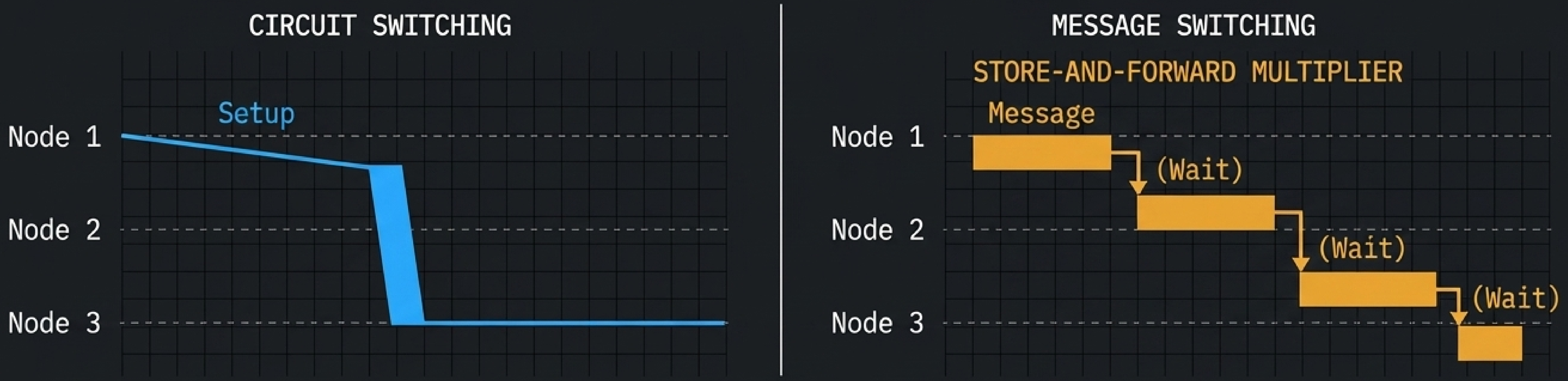
ENGINEER'S LOG

Space-division switching utilizes massive crossbars, but N^2 crosspoint scaling is unsustainable, leading to the invention of multi-stage Clos Networks.

Time-Division switching moves the problem from wire to memory. Using a TSI, we write frames sequentially to RAM and read them out in a mapped order. Throughput is hard-capped by memory access speed.

MESSAGE SWITCHING & THE STORE-AND-FORWARD PENALTY

Grasp the severe latency and memory limitations of whole-message store-and-forward routing.



Metric	Circuit Switching	Message Switching
Forwarding Model	Contiguous Bitstream	Store-and-Forward
Switch Memory Requirement	Minimal (fabric state only)	Massive (entire message payload)
Error Handling Penalty	Bit-level recovery	Must dump and retransmit whole message
Transmission Time (L/R)	Occurs exactly once	Multiplied by number of hops

ENGINEER'S LOG Message switching introduced the store-and-forward paradigm. Look at the timing diagram: an intermediate switch must receive the entire* message, store it to disk, compute the checksum, and only then begin forwarding. This creates the "staircase of death" where the transmission delay is multiplied by every single hop. This fragility forced the industry to invent fragmentation.

DATAGRAMS VS. VIRTUAL CIRCUITS

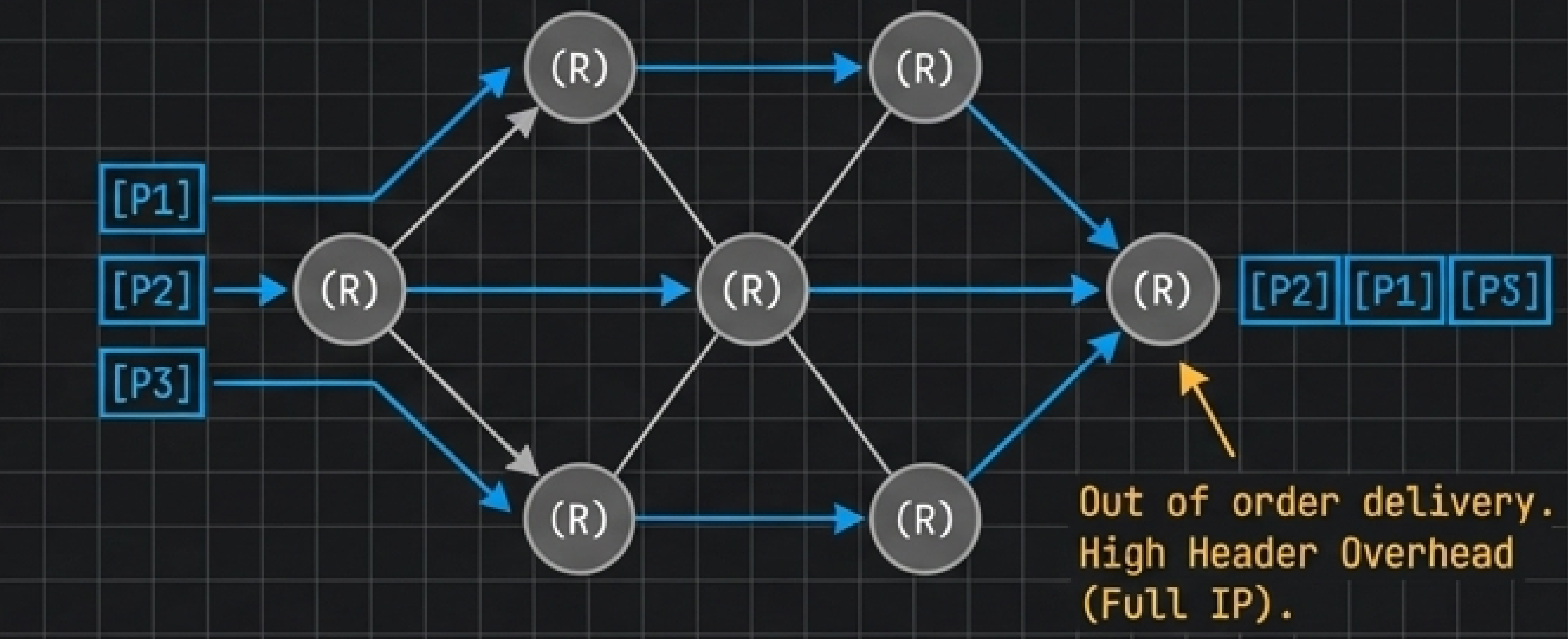
Contrast connectionless and connection-oriented packet switching architectures.

Feature	Datagram (IP)	Virtual Circuit (ATM)
Path Determination	Independent per packet	Fixed at setup phase
Header Overhead	High (Full source/dest IP)	Low (Short VCI/VPI tag)
Failure Resilience	High (dynamically reroutes)	Low (path breaks, setup fails)
Packet Ordering	Out of order possible	Guaranteed sequential delivery

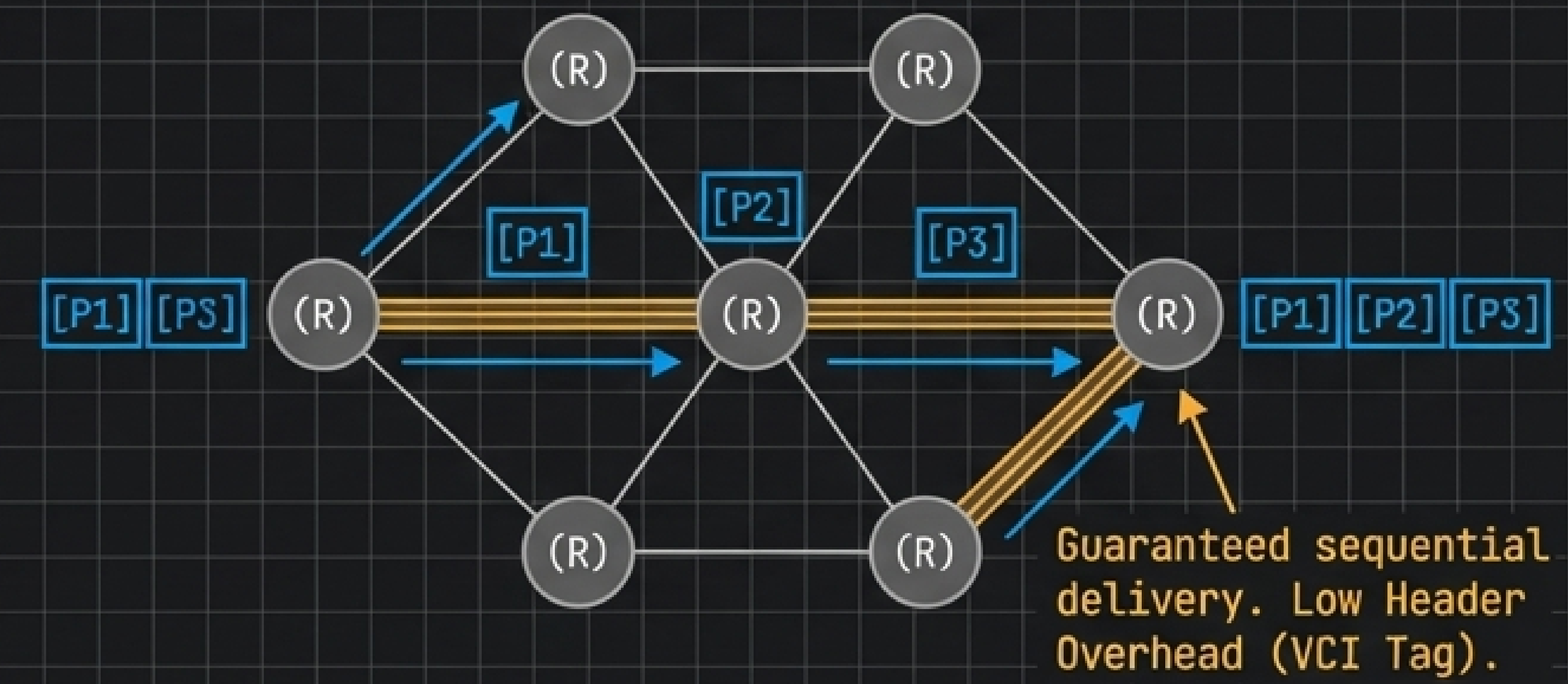
ENGINEER'S LOG

By fragmenting data, we pipeline transmission, overlapping delays. Datagram networks route each packet independently—highly resilient, but prone to out-of-order delivery. Virtual Circuits borrow from circuit switching, mandating a setup phase to lock a logical path. This allows subsequent packets to carry tiny VCI tags instead of heavy headers. It's a trade-off between edge-node complexity and core-network efficiency.

DATAGRAM (Connectionless)



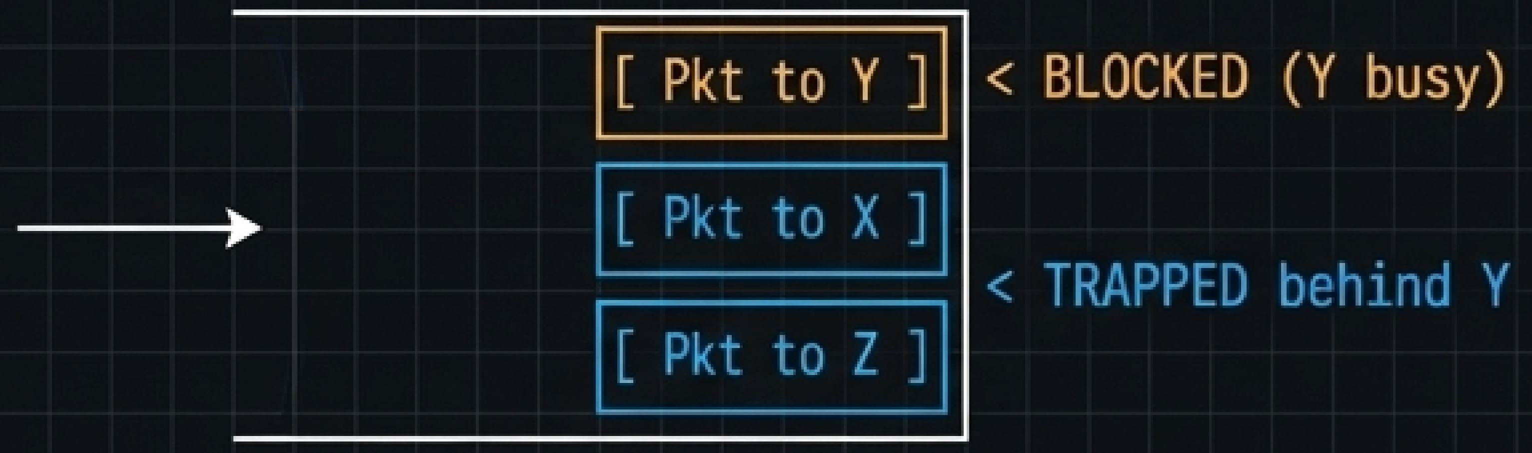
VIRTUAL CIRCUIT (Connection-Oriented)



THE FABRIC BOTTLENECK: HEAD-OF-LINE BLOCKING

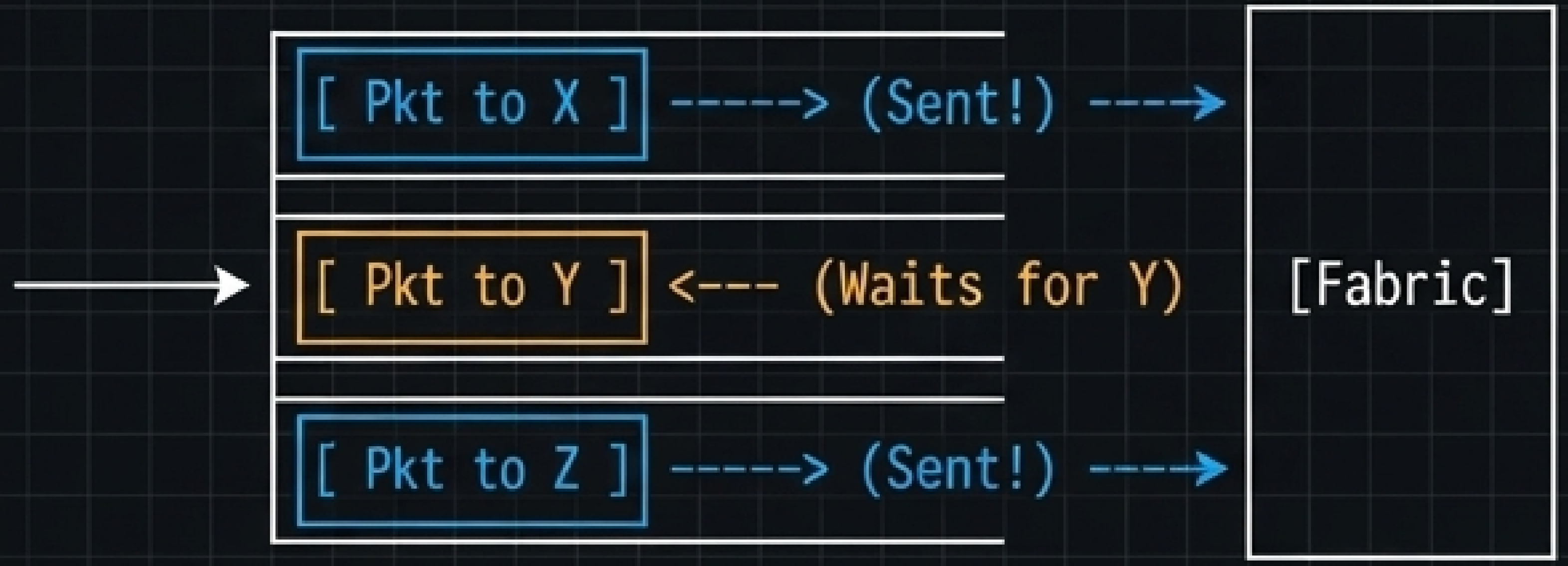
Understand how VOQ eliminates Head-of-Line blocking to maximize switch fabric throughput.

HEAD-OF-LINE (HOL) BLOCKING



Result: Max ~58.6% throughput (Math proven)

VIRTUAL OUTPUT QUEUING (VOQ)



Result: ~100% theoretical throughput

ENGINEER'S LOG

In a basic input-queued switch, a blocked packet at the front traps completely independent packets behind it. VOQ fixes this by maintaining separate logical FIFOs for every output port inside the input buffer, achieving nearly 100% fabric utilization.

Queuing Model	Architecture	Blocking Mechanic	Max Throughput
Input Queuing (FIFO)	Single line per input port	Front packet traps rest	~58.6%
Output Queuing	Buffers exist at output	Requires N x fabric speed	100% (Hardware expensive)
Virtual Output Queuing	Logical FIFOs per output	Decouples destinations	~100% (Standard today)

PARADIGM SYNTHESIS: THE SWITCHING TRILEMMA

Synthesize the core characteristics of Circuit, Datagram, and VC switching to select the optimal architecture

Paradigm	Path Dedication	Bandwidth Allocation	Congestion Management
Circuit Switching	Physical Path	Reserved / Guaranteed	Blocked at setup
Datagram Packet (IP)	None (Fluid)	Dynamic / Best Effort	Packets dropped
Virtual Circuit	Logical Path	Reserved / Managed	Negotiated at setup

Feature	Circuit Switching	Datagram Packet (IP)	Virtual Circuit (ATM)
Dedicated Path	Yes (Physical)	No	Yes (Logical)
Setup Phase	Required	None	Required
Header Overhead	None (after setup)	High (Full IP address)	Low (Short VC tag)
Congestion Failure	Busy signal at setup	Packets dropped in transit	Handled via queues

ENGINEER'S LOG

Notice how Virtual Circuit switching attempts to thread the needle. It steals the connection-oriented predictability of circuit switching to minimize header overhead, but keeps the multiplexed packetization of datagrams to share the wire. The physical layer underneath all of these is still a shared wire.

MULTIPLEXING TAXONOMY: DIVIDING THE PHYSICS

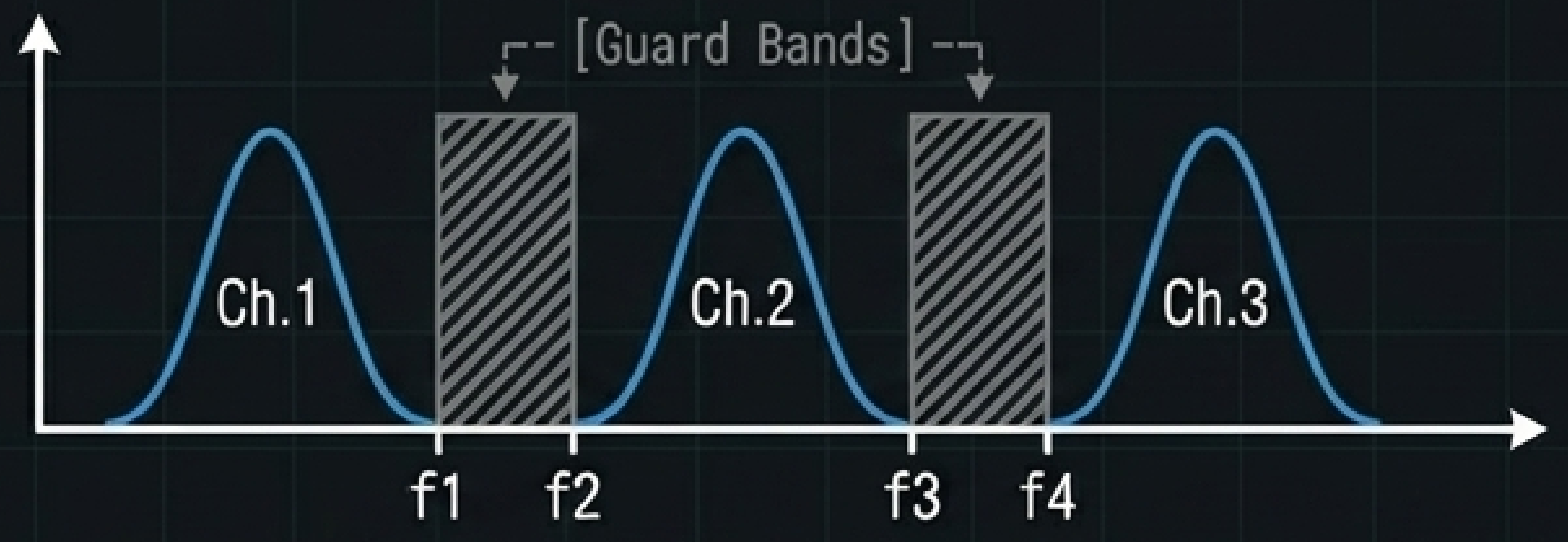
Differentiate how FDM and WDM partition analog bandwidth using frequency and wavelength separation.

Technology	Medium	Separation Metric	Isolation Mechanism
FDM	Copper / Airwaves	RF Frequency (Hz)	Guard Bands (empty spectrum)
WDM	Fiber Optic	Wavelength (nm/color)	Optical Prisms / Filters
DWDM	Fiber Optic	Dense Wavelengths	Extreme precision lasers (160+ channels)

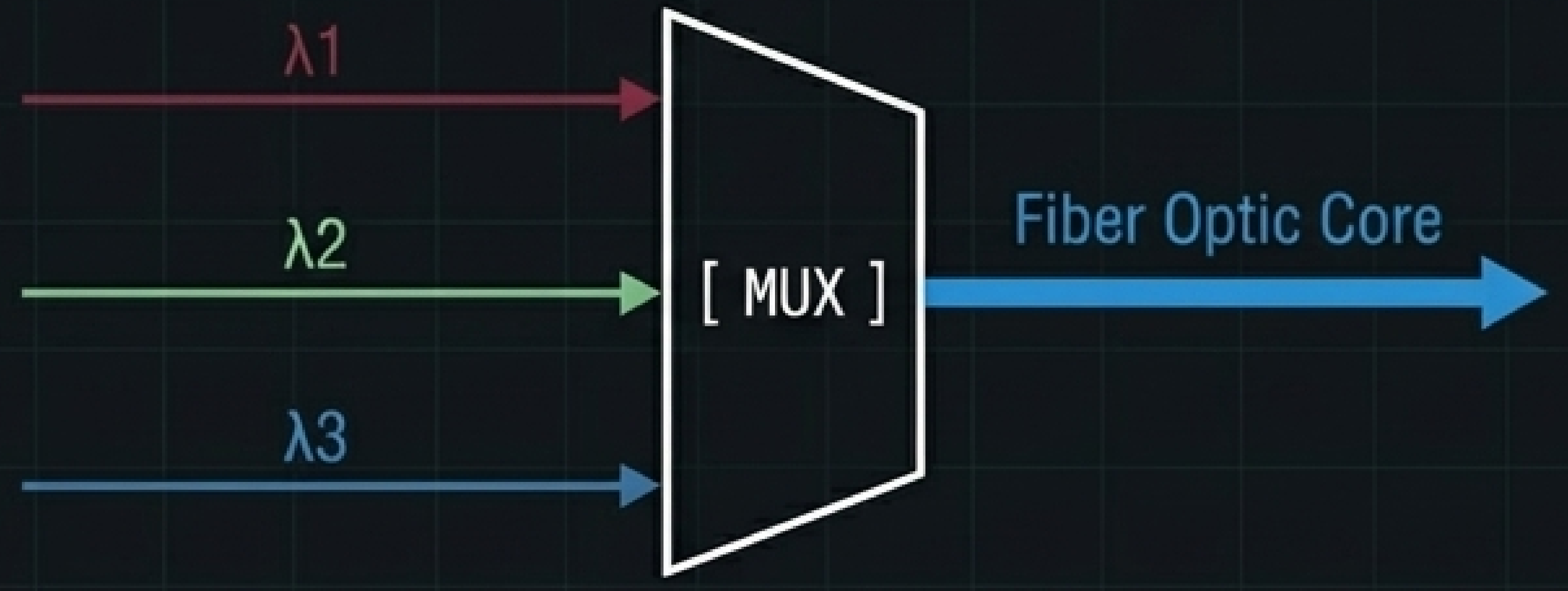
ENGINEER'S LOG

Multiplexing is the physical layer's answer to resource sharing. Frequency Division Multiplexing divides spectrum into distinct channels, intentionally wasting space with Guard Bands to prevent crosstalk.
Wavelength Division Multiplexing is the optical equivalent. Using highly calibrated lasers, Dense WDM injects over 160 distinct wavelengths of light down a single strand of glass. In both cases, the division is analog and permanent.

FREQUENCY DIVISION (FDM)



WAVELENGTH DIVISION (WDM/DWDM)

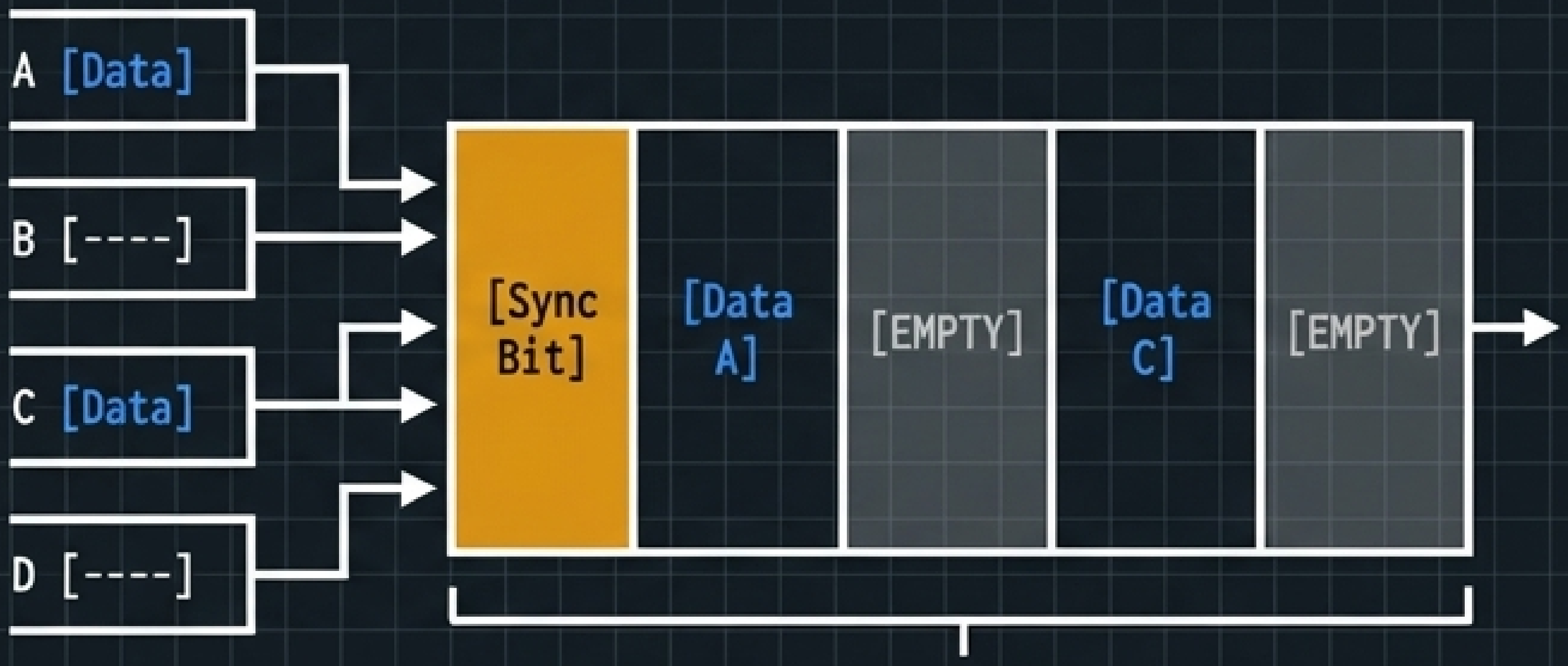


Time-Domain Slicing

Master the mechanics of Synchronous Time Division Multiplexing (STDM) and framing synchronization.

CONCEPT COMPARISON			
Concept	FDM Counterpart	TDM Implementation	Purpose
Resource Slicing	Frequency Band	Time Slot	Divide physical capacity
Interference Prevention	Guard Band (Hz)	Guard Time (microsec)	Absorb drift/ propagation delay
Demux Alignment	Bandpass Filter	Framing / Sync Bits	Identify start of sequence

STDM FRAME INTERLEAVING (Round Robin)



Strict temporal rotation. Position implies identity.

ENGINEER'S LOG

STDM slices the wire vertically by time, dividing the timeline into fixed frames and statically assigning a slot to each input. Because this is a continuous bitstream, the receiver has no idea where a frame begins. We must inject a "framing bit" at the start of every frame. We also use microsecond silences between slots—Guard Times—analogous to Guard Bands in FDM.

The Efficiency Dilemma of STDM

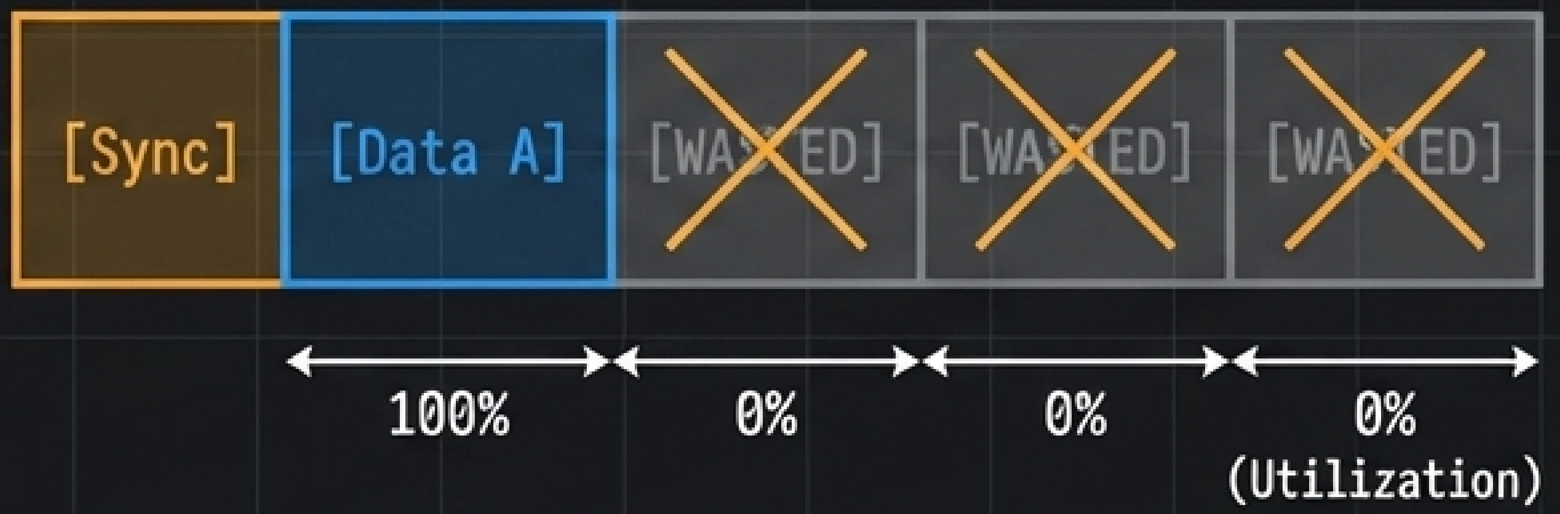
Calculate network capacity requirements and identify the inherent bandwidth waste of static slot allocation.

STDM LINK CAPACITY REQUIREMENT

$$C = (N * R) + R_{overhead}$$

N = Number of inputs
R = Max bit rate per input

WASTE CONDITION



Inputs B, C, and D are idle.

Metric	STDM Realities	Impact on Network
Link Provisioning	C >= Sum(R_peak)	Massive physical capacity required
Active Source Yield	Slot transmits payload	High efficiency per active slot
Idle Source Yield	Slot remains completely blank	Catastrophic bandwidth waste
Queuing Delay	Strictly Zero	Predictable, perfect for voice PSTN

ENGINEER'S LOG

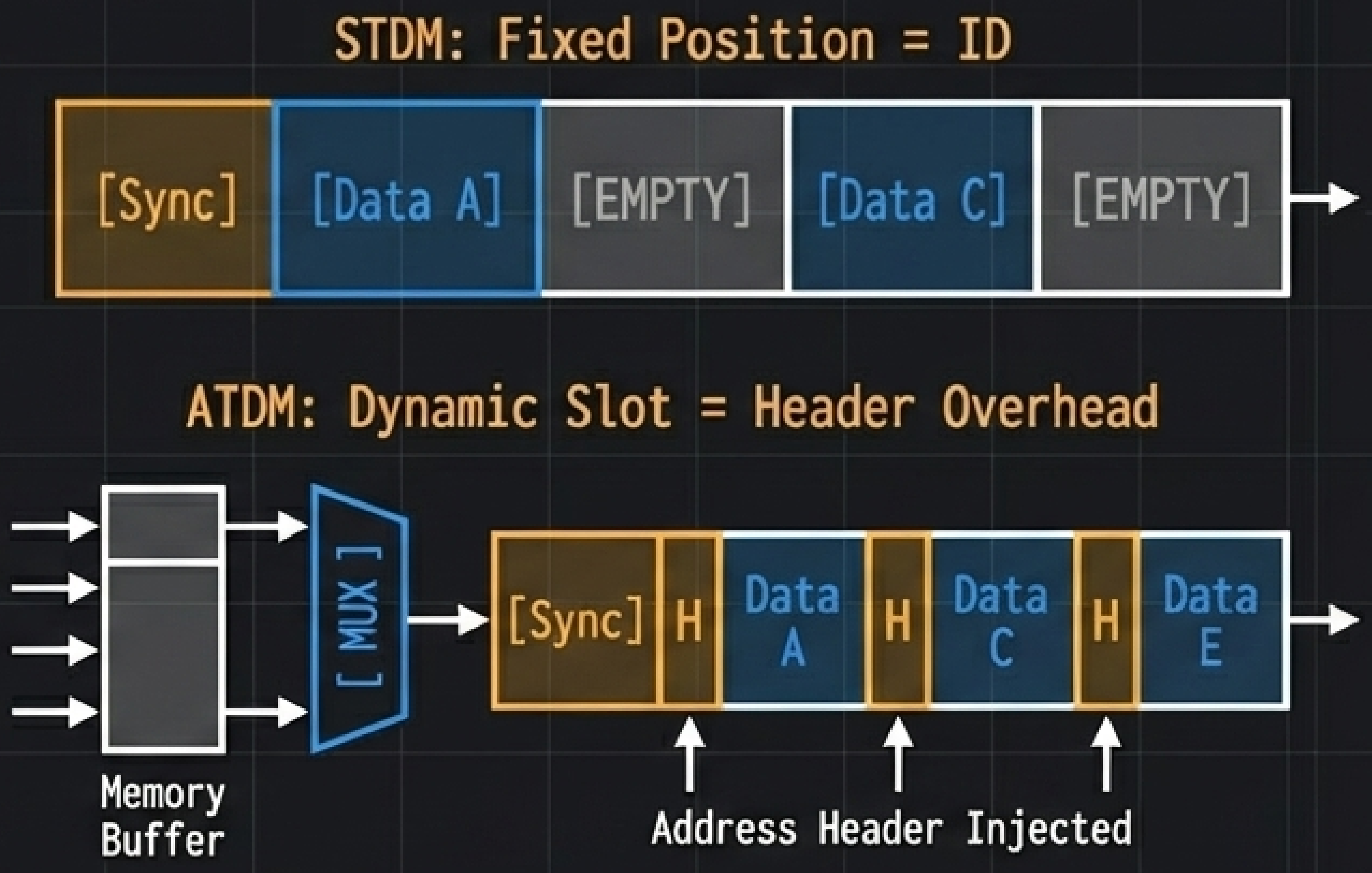
The math of STDM reveals a fatal flaw. If you multiplex N sources at rate R, the output link must permanently support N*R. If a data source has nothing to send, its time slot rolls past completely empty. It is mathematically hostile to the bursty nature of computer data.

Statistical TDM (ATDM): Dynamic Allocation

Understand how dynamic slot allocation achieves near 100% utilization at the cost of addressing overhead and jitter.

Feature	Synchronous TDM (STDM)	Statistical TDM (ATDM)
Slot Allocation	Fixed, round-robin	Dynamic, strictly demand-driven
Addressing Overhead	None (position implies ID)	Required in every payload slot
Bandwidth Utilization	Low (wasted idle slots)	High (near 100% efficiency)
Delay Profile	Fixed, zero queuing delay	Variable (jitter), queues build up

ENGINEER'S LOG
To fix STDM's wasted bandwidth, we adopt Statistical TDM. The MUX scans input lines and only collects data from active sources, packing the frame densely. The trade-off? Since a slot's position no longer implies its owner, every single payload must carry an addressing header. We trade bandwidth efficiency for header overhead and variable queuing delay.



Statistical Multiplexing Gain & Queuing Theory

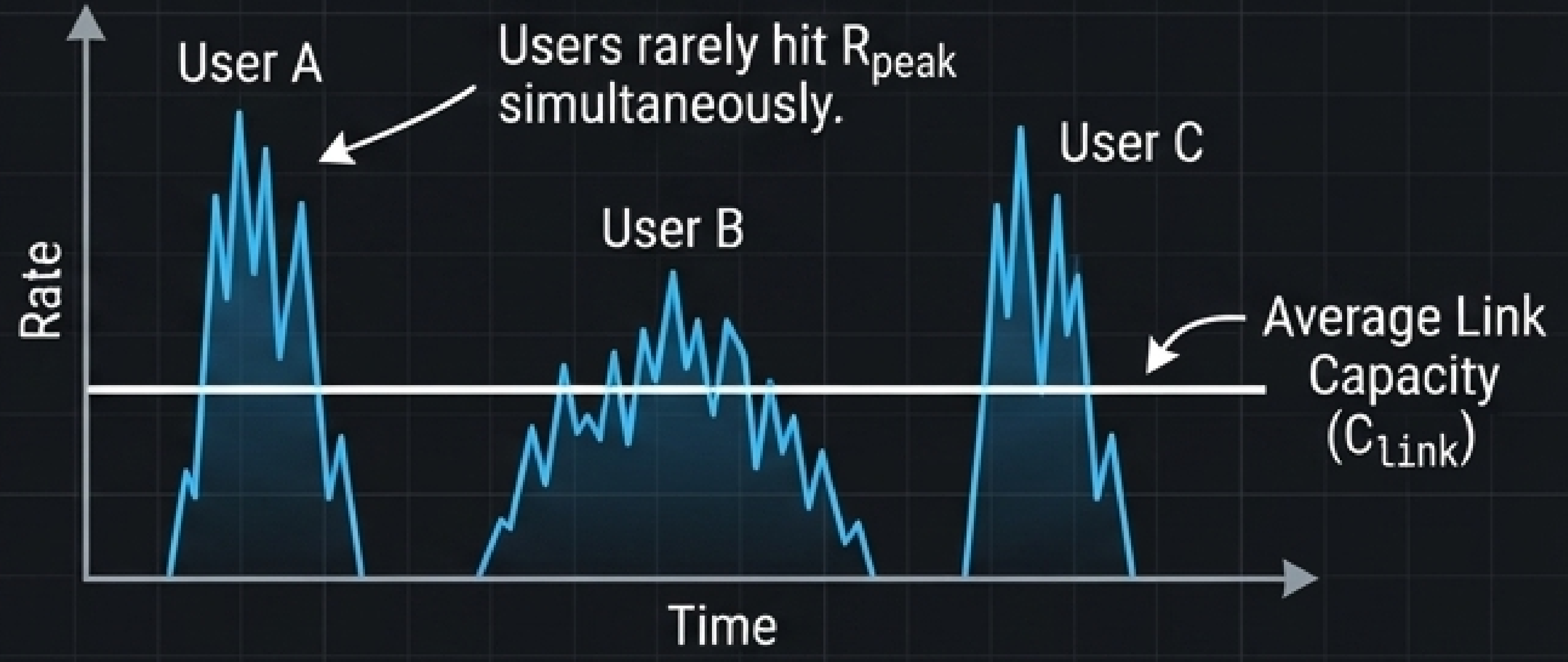
Learn how ISPs use bursty traffic profiles to mathematically oversubscribe network links.

SMG & QUEUING MATH

$$SMG = \frac{Sum(R_{peak})}{C_{link}} \qquad \rho\ (\rho) = \frac{\lambda}{\mu}$$

If p >= 1, Queue length approaches Infinity.

TRAFFIC PROBABILITY GRAPH



ENGINEER'S LOG

Here is the economic engine of modern ISPs. Because internet traffic is heavily bursty, we can safely oversubscribe the hardware.

An SMG greater than 1 means oversubscription.

However, queuing theory dictates that if arrival rate exceeds service rate, utilization hits 1. At that exact moment, queues grow toward infinity, and the switch begins violently dropping packets.

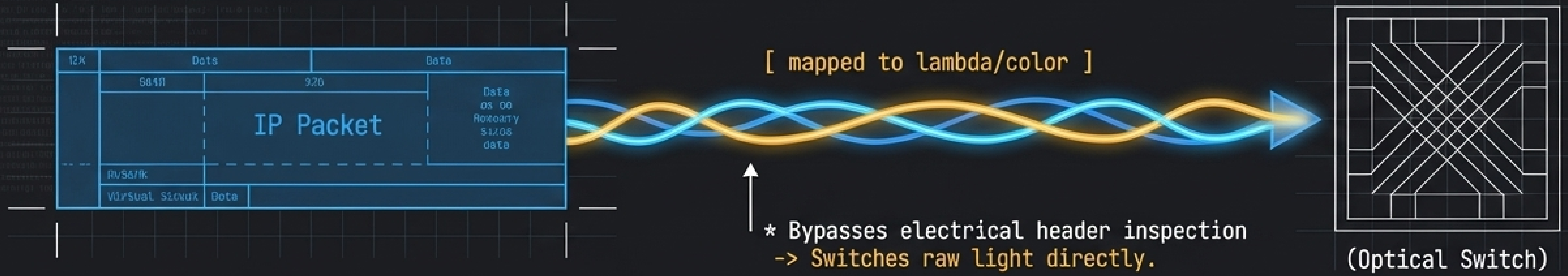
Traffic Type	Peak vs Average Ratio	SMG Feasibility	Risk of Queue Overload (p>=1)
Voice (Uncompressed)	1:1 (Constant Rate)	SMG = 1 (No gain)	Zero
Video Streaming	2:1 (Variable Rate)	Low SMG	Low
Web / Data LAN	10:1 to 100:1 (Bursty)	High SMG (>10)	High (requires deep buffers)

The Modern Convergence

Learning: Synthesize switching and multiplexing history by examining the ATM cell compromise and the rise of optical lambda-switching.

ATM CELL COMPROMISE (Voice/Data VC Packet)

OPTICAL CONVERGENCE (GMPLS)



TECHNOLOGY MILESTONE	PRIMARY ARCHITECTURE	THE GENIUS INNOVATION	CORE COMPROMISE / LIMITATION
SONET / SDH	Synchronous TDM	Carrier-grade clock synchronization	Rigid capacity mapping
ATM	Virtual Circuit Packet	53-byte cell (Voice/Data hybrid)	High header overhead (~10%)
GMPLS	Circuit + WDM hybrid	lambda-switching (Routing light)	Massive optical complexity

ENGINEER'S LOG

Why ATM's bizarre 53-byte cell? Voice requires tiny payloads to prevent delay, while data wants massive payloads to dilute overhead. They compromised at 48+5.

Today, GMPLS completely transcends this. Instead of inspecting packet headers in silicon, we route entire wavelengths of light natively. It blurs the line between circuit switching, routing, and WDM, bringing us full circle.